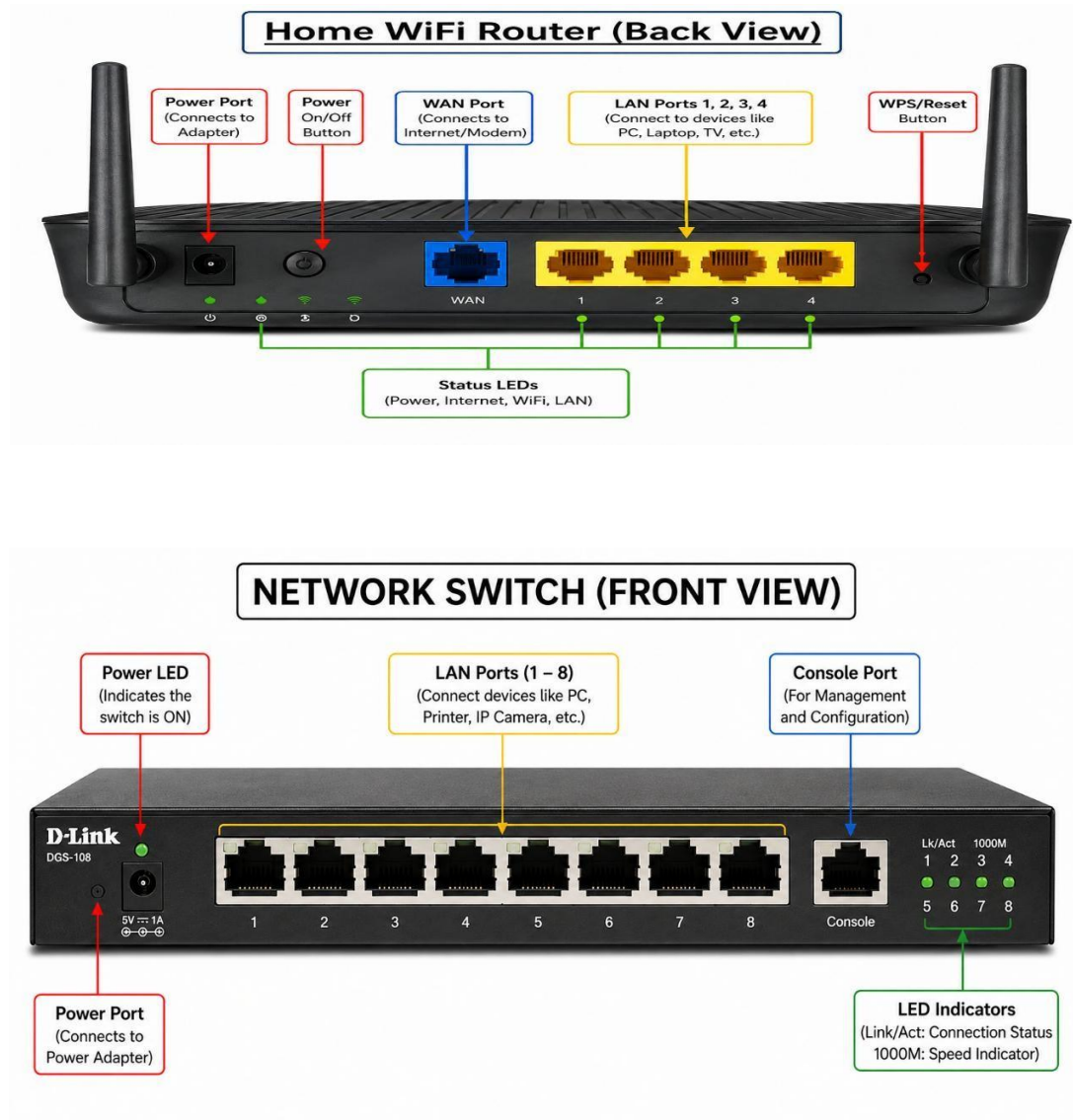


1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.



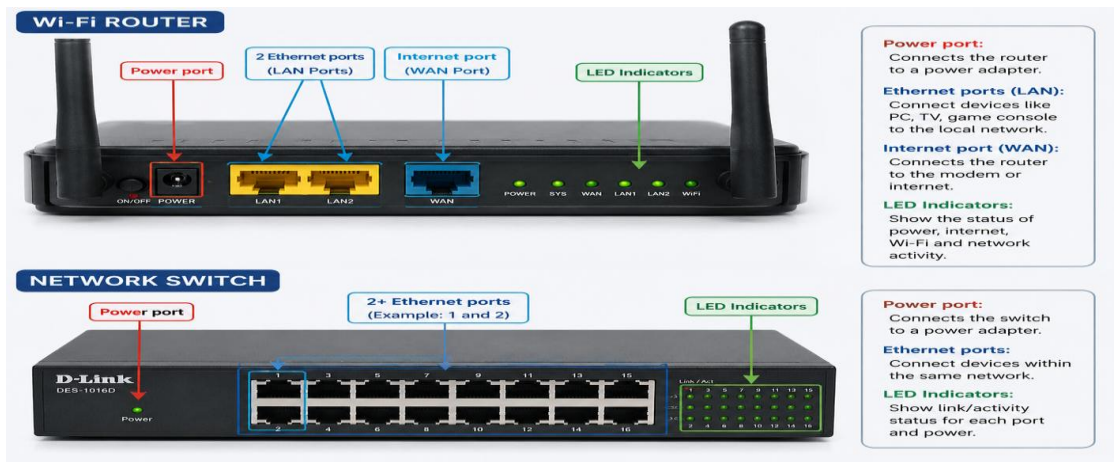
2. List 3 key differences between routers and switches in terms of their main functions and how they handle

network traffic.

Hint:
Think about how each device deals with IP addresses,
MAC addresses, and the types of networks they
connect.

Feature	Router	Switch
Main Function	Connects different networks, such as a home network to the Internet.	Connects multiple devices within the same Local Area Network (LAN).
Address Used	Uses IP addresses (Layer 3) to route data between networks.	Uses MAC addresses (Layer 2) to forward data to the correct device.
Network Traffic Handling	Directs traffic between different networks and provides features like NAT, DHCP, and firewall.	Forwards data only within the local network, making communication between connected devices fast and efficient.

3. Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.



Component	Position	Purpose
Power Port	Back side, left corner	Connects the router to the power adapter.
WAN Port	Blue Ethernet port on the back	Connects the router to the modem or ISP Internet line.
LAN Port 1	First yellow Ethernet port	Connects wired devices such as a PC.
LAN Port 2	Second yellow Ethernet port	Connects another wired device like a printer or TV.
LED Indicators	Front panel	Show Power, Internet, Wi-Fi, and LAN connection status.

B. Network Switch

Component	Position	Purpose
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Power Port	Left side/back	Connects the switch to the power adapter.
Ethernet Port 1	First RJ-45 port	Connects a computer or another network device.
Ethernet Port 2	Second RJ-45 port	Connects another PC, printer, or IP camera.
LED Indicators	Above or beside Ethernet ports	Show power status and network activity (Link/Act).

4. Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Scenario:

I am setting up a small gaming café with **8 gaming PCs** for IPL live streaming and multiplayer games.

First, I would connect the **router** directly to the **ISP (Internet Service Provider) line** because the router receives the Internet connection, assigns IP addresses using DHCP, and provides security through NAT and a firewall.

Next, I would connect an **8-port network switch** to one of the router's LAN ports using an Ethernet cable.

Finally, I would connect all **8 gaming PCs** to the switch using Cat6 Ethernet cables.

Justification

- The **router** is required to connect to the Internet and manage network traffic.
- The **switch** efficiently connects multiple computers within the same local network.

- Using wired Ethernet connections provides **low latency**, **high-speed data transfer**, and a **stable connection**, which are essential for online multiplayer gaming and IPL HD streaming.
- This setup ensures all 8 gaming PCs can access the Internet while communicating efficiently over the local network.